SAMAR ANAND

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Education

VIT Bhopal University *Bhopal, Madhya Pradesh*
B.Tech in Computer Science (Artificial Intelligence and Machine Learning) Aug 2023 – Aug 2027
CGPA: 8.63/10.00

Technical Skills

Languages: C++, JavaScript, TypeScript, SQL
Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems
Backend: Node.js, Express.js, FastAPI, REST APIs
Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS
Databases & Tools: MongoDB, PostgreSQL, MySQL, Git, GitHub, Docker, AWS, Vercel, Render
AI/ML: TF-IDF, Scikit-learn, RAG, LangChain, LangGraph, Hugging Face

Projects

VibeCode — AI-Powered Web IDE | *Next.js 15, TypeScript, WebContainers, Monaco Editor, Prisma, MongoDB* | GitHub | Demo
- Built a browser-based AI development environment using Next.js 15 and WebContainers, supporting code execution across React, Next.js, Express, Hono, Vue, and Angular projects.
- Integrated Google Gemini API with a local Ollama fallback to provide AI-powered code completion and context-aware project refactoring assistance.
- Implemented Google and GitHub OAuth authentication using NextAuth v5 and persisted user workspaces and project data with Prisma and MongoDB.
- Built an interactive terminal and file explorer using xterm.js and WebContainers, supporting browser-based file creation, editing, and project execution without local setup.

MedQueue AI — Real-Time Triage & Priority Queue | *React 18, Node.js, Express, Socket.io, Zod, MongoDB* | GitHub | Demo
- Built a real-time clinical intake platform using React, Node.js, Express, Socket.io, and MongoDB to manage patient intake, triage, and queue prioritization.
- Implemented a local rule-based fallback engine that maintains triage functionality when external AI APIs encounter HTTP 429 rate limits or network failures.
- Implemented real-time queue prioritization using MongoDB transactions, allowing higher-priority ESI 1–4 patients to automatically move ahead in the queue.
- Developed bi-directional Socket.io communication across Nurse Station, Doctor Workbench, and ER Lobby dashboards, with the frontend deployed on Vercel and backend on Render.

CartPilot — Multi-Agent Grocery Assistant | *LangGraph, Next.js 14, TypeScript, Express, MongoDB Atlas, SSE* | GitHub | Demo
- Built an autonomous multi-agent grocery shopping platform using LangGraph, Next.js 14, and TypeScript, orchestrating a 5-agent pipeline (Planner, Researcher, Recommender, Cart Manager, Validator) to assemble constraint-based grocery carts.
- Implemented a self-healing replanning cycle with LangGraph conditional routing, dynamically adjusting product quantities and replacing high-cost items to satisfy deterministic budget caps and live inventory limits.
- Integrated multi-city fulfillment across 20 Indian dark stores by querying real-time MongoDB catalog stock and localized pricing based on delivery pincodes, with resilient multi-tier query fallbacks.
- Streamed real-time agent execution events to the frontend via Server-Sent Events (SSE), providing a live workflow timeline and an interactive proposal approval checkpoint before cart transfer.

Experience

Ethnus May 2025 – Jul 2025
MERN Stack Intern *Remote*
- Developed RESTful backend services for core platform modules.
- Optimized MongoDB schemas and queries, improving data retrieval speed and database efficiency.
- Developed responsive React interfaces across key application pages, enhancing user experience and rendering performance.
- Implemented authentication and role-based access control (RBAC) to secure application endpoints and user roles.

ServiceNow May 2026 – Jun 2026
Virtual Internship Program — ServiceNow University, AICTE & SmartBridge *Remote*
- Worked with ServiceNow administration concepts including user management, platform configuration, workflow automation, reporting, and dashboard creation.
- Explored Agentic AI capabilities and automated workflow development within the ServiceNow platform.
- Tested custom application logic using ServiceNow Automated Test Framework (ATF) configurations.

Certifications & Achievements

- **Achievement:** Selected as a mentee among 60,000+ participants in Algo University camp. 5+ Hackathons — 2× Adobe Hackathon
- **Leadership:** Appointed as Campus Ambassador for Internsala student program and GSSoC.
- **Certifications:** Microsoft Azure Fundamentals DP-900 (Microsoft), MERN Full Stack Development (Ethnus), Applied Machine Learning in Python (University of Michigan).